MIL-B-60089B (MU)
17 June 1969
SUPERSEDING
MIL-B-60089A (MU)
4 March 1968

MILITARY SPECIFICATION

BURSTER, CHEMICAL AGENT MINE, M38; INITIATOR,

MINE BURSTER, M48

1. SCOPE

1.1 This specification covers some of the explosive elements pertinent to the M23 chemical agent mine.

2. APPLICABLE DOCUMENTS

2.1 Government documents: The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extend specified herein:

SPECIFICATIONS

FEDERAL

O-A-51 - Acetone, Technical.

MILITARY

MIL-G-155 - Graphite, Dry (For Use in Ammunition).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

FSC 1345

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C37-1-63	-	Burster, Gas Mine, M38 (Assembly, Detail and Bill of Material).
C37-1-64	-	Initiator, Mine Burster, M48, (Assembly, Detail and Bill of Material).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179	-	Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179".)

3. REQUIREMENTS

3.1 Materials. All materials and quantities cited on Drawings C37-1-63 and C37-1-64 or on subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.2.1 Tetryl pellet composition. The tetryl pellets used in the booster (Drawing C37-1-63) may include graphite (MIL-G-155, grade I, II, III or IV) to facilitate manufacture. However, the graphite (see 6.2) shall be uniformly blended with the tetryl and shall not exceed 2 percent of the pellet when tested in accordance with 4.2.4.1.

3.2.2 Burster. The burster shall be loaded with a uniform composition of the explosive mixture specified on Drawing C37-1-63. The loaded burster without tetryl booster and aluminum tape shall weigh at least 390 grams (g), when tested in accordance with 4.2.4.2.

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3.2.3 Initiator. The initiator shall be loaded with a uniform composition of the explosive mixture specified on Drawing C37-1-64. The loaded initiator shall weigh at least 55 g when tested in accordance with 4.2.4.2.

3.3 Workmanship. The explosive elements shall be free from foreign matter such as oil and grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting.

4.2.1.1 Bursters. A lot shall consist of the bursters of one interfix number, produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one batch of composition B4 shall be represented in any one lot of bursters (see 6.3).

4.2.1.2 Tetryl pellets. A lot shall consist of the applicable size pellets of one interfix number, produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of tetryl shall be represented in any one lot of finished pellets. If binders and

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lubricants are used, no more than one batch (see 6.3) shall be represented in any one lot of tetryl pellets.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For nondestructive tests, level S-4 shall be used.

4.2.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-2.

4.2.3 Inspection procedures.

4.2.3.1 For examination and nondestructive tests. Sample elements shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For destructive tests. Sample pellets shall be tested in accordance with 4.2.4.1 and with MIL-STD-105 using an AQL of 1 percent defective.

4.2.3.3 Classification of defects.

(a) Burster, gas mine, M38 assembly (Drawing C37-1-63).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Height of filling incorrect	
102	Dimensions of cavity in filling incorrect	
103	Location of cavity in fill incorrect	
104	Weight of burster incorrect	4.2.4.2
105	Tetryl pellet missing	
106	Tetryl pellet dimensions incorrect	
107	Aluminum tape missing	
108	Workmanship (3.3)	

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4.2.3.3 Classification of defects (continued)

(b) Initiator, mine burster, M48-assembly (Drawing C37-1-64).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Relation of explosive to end of tube incorrect	
102	Weight incorrect	4.2.4.2
103	Workmanship (3.3)	

(c) Tube, charge (Drawing A37-1-17).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Camber excessive
102	Length incorrect
103	Diameter incorrect
Minor:	AQL 4.0 percent defective
201	Protective finish missing
202	Burrs

4.2.4 Tests.

4.2.4.1 Tetryl pellet composition. A composite sample of approximately 10 grams of tetryl pellets shall be separately weighed to the nearest 0.5 milligram tared extraction thimble. A soxhlet extraction apparatus containing approximately 50 milliliters (ml) of acetone conforming to O-A-51 shall be used. Heat shall be applied to the apparatus until a constant rate of reflux is reached, the rate being at least 60 drops per minute. The maximum reflux rate shall be such that the liquid level remains at least 16 ml below the top of the thimble. The amount of solvent in the apparatus shall always be sufficient to keep the tetryl in solution. The extraction shall continue for a minimum period of 2 hours and for 1 hour

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after the filtrate becomes colorless. Upon completion of the extraction, the thimble shall be placed in a well-ventilated drying chamber and dried to a constant weight at a temperature of 50° ± 2°C. The residue shall be calculated as the percent of binders, lubricants and inert material in the pellets.

4.2.4.2 Weight. The weight of the burster and the initiator shall be determined on a beam scale whose accuracy is within 1.0 g.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. Bursters and initiators shall be packaged separately and in such a manner as to afford adequate protection against corrosion, deterioration and damage when shipped from the supply source to the first receiving destination.

5.2 Packing, level C. Bursters and initiators packaged in accordance with 5.1 shall be packed separately to insure carrier acceptance and safe delivery to the first domestic destination. Containers and weight limitations shall be in compliance with DOT regulations.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 and DOT regulations.

6. NOTES

6.1 Ordering data. Procurement documents should specifiy the title, number and date of this specification.

6.2 Lubricant. The use of at least 0.25 percent of graphite is recommended when forming tetryl pellets.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some unit physical or chemical mixing process intended to yield a substantially uniform product.

6.4 Precautions. Because of the hazards attending the manufacture and assembly of explosives, suppliers are advised to exercise all necessary precautions to protect personnel from explosions during manufacture and handling of these items.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1345-A198

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply *authorization to waive any portion of the* referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 55 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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